THE VIRTUAL MEET EXPERIENCE

2017 - 2018

HS VIRTUAL CHALLENGE MEET #1

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge Meet #1 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is $64_{10} - 11111_2$?

A. 32_{10} B. 100001_2 C. 31_8 D. 22_{16} E. None are equivalent

QUESTION 2

What is output by the code to the right?

A. 17 B. 15

C. 16 D. 14

E. There is no output due to a compile error.

```
int k=9;
k+=6.2;
System.out.println(k);
```

QUESTION 3

What is output by the code to the right(e's indicate blank spaces)?

A. 03.60eefunzoeee6 B. e3.60eefunzoeee6

C. 03.6000funzo0006 D. e3.60efunzoee6

E. 03.60funzo6

```
System.out.printf
      ("%05.2f%7s%4d",3.6,"funzo",6);
```

QUESTION 4

What is output by the code to the right?

A. JebronLn B. JebronLameso

C. JebronLo D. JebronLamesn

E. There is no output due to a syntax error.

```
String k="JebronLames";
k.substring(0,7);
k+=k.charAt(5);
System.out.println(k);
```

QUESTION 5

What is output by the code to the right?

A. True B. False

C. false D. true

E. There is no output due to a syntax error.

```
boolean t=true;
boolean f=false;
System.out.println
                 (t^t&&f||t&&t&&f^t);
```

QUESTION 6

What is output by the code to the right?

A. 6.0 B. 6

C. 5.0 D. 5

E. There is no output due to a syntax error.

```
int y=Math.round(5.45);
System.out.println(y);
```

QUESTION 7

What is the output by the code to the right?

A. 28 B. 29

C. 1210 D. 129

E. 210

```
int y=8;
if(y<7 && ++y==9)
     System.out.print(1);
if(y>2 || ++y==10)
     System.out.print(2);
System.out.println(y);
```

QUESTION 8

What is the output by the code to the right?

A. 9 B. 8
C. 7 D. 6
E. 5

```
System.out.println(8*7/5+3%2-6);
```

QUESTION 9

What is the output by the code to the right ?

A. 5354555657 B. 1424344454
C. 56789 D. 5253545556
E. There is no output due to a syntax error.

```
String c="12345";
for(char k:c.toCharArray())
   System.out.print(k+4);
```

QUESTION 10

What is the output by the code to the right ?

A. a0A K B. 9748653275
C. There is no output due to a syntax error.
D. There is no output due to a runtime error.
E. Output cannot be determined until runtime.

```
char[] c=new char[]
              {'a','0','A',' ','K'};
System.out.println(c);
```

QUESTION 11

What is output by the code to the right if the input is the sequence shown below?

3 -1 4 Samuel 9 #@$ -1

A. 3 B. -1Samuel#@$
C. -1
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
Scanner file=new Scanner(System.in);
while
  (file.hasNext()&&file.nextInt()>0)
     System.out.print(file.next());
```

QUESTION 12

What is the output by the code to the right ?

A. 512 B. 256
C. 128 D. 64
E. 0

```
int y=1;
for(int k=1;k<8;k++)
    y*=2;
System.out.println(y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. III, IV, II, I B. IV, III, II, I
C. I, II, III, IV D. II, IV, III, I
E. IV, III, II, I

```
I. ?:
II. ++
III. instanceof
IV. ~
```

QUESTION 14

What is the output by the code to the right ?

A. 4 B. 64
C. 8 D. 32
E. 16

```
System.out.println(Short.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. `[5, 4, 3]`
B. `[3, 2]`
C. `[5, 4, 2]`
D. `[5, 4]`
E. There is no output due to a syntax error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
a.add(5);a.add(4);a.add(3);
a.add(2);a.remove(2);
System.out.println(a);
```

QUESTION 16

What is the output by the code to the right ?

A. `13`
B. `14`
C. `16`
D. `15`
E. There is no output due to a syntax error.

```
System.out.print(5|6|7|8&9);
```

QUESTION 17

What is the output by the code to the right ?

A. `True`
B. `False`
C. `true`
D. `false`
E. There is no output due to an error.

```
String h="funzo";
String t=h;
System.out.println
      ("Are they Equals? " + h == t);
```

QUESTION 18

What could replace **`<1*>`** in the code to the right, so that s returns the value that is removed?

A. `poll`
B. `pop`
C. `remove`
D. `kill`
E. More than one of the choices above is correct.

QUESTION 19

What is the output by the code to the right ?

A. `20`
B. `18`
C. `21`
D. `19`
E. There is no output due to a syntax error.

```
Stack<Integer> s;
s=new Stack<Integer>();
int k=0;
s.push(3);s.push(8);
s.push(7);s.push(4);
k+=s.<1*>()+s.<1*>();
s.push(5);s.push(9);
System.out.println(k+s.<1*>());
```

QUESTION 20

Which of the following is a legal instantiation?

A. `Map<String>v=new TreeMap<String,Integer>();`
B. `List<Integer>f=new TreeSet<Integer>();`
C. `List<Integer>l=new Queue<Integer>();`
D. `Object c=new Comparable();`
E. None of the above are legal.

QUESTION 21

What is the output by the code to the right ?

A. `FunzoABCaA1H`
B. `fUNZOabcAa!h`
C. Output cannot be determined until runtime.
D. There is no output due to syntax error
E. There is no output due to a runtime error.

```
String s="Funzo";
char[] c=new char[]{'a','A','1','H'};
System.out.println(s+"ABC"+c);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right?

A. `super(a,b);`
B. `super();`
C. `y=a;x=b;`
D. A and B
E. A and C

QUESTION 23

Which of the following could replace **<2*>** in the code to the right?

A. `A();`
B. `A(5,7);`
C. `B();`
D. `B(5);`
E. More than one of the above.

QUESTION 24

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by `// line 1` ?

A. `48`
B. `36`
C. `24`
D. `12`
E. This is no output due to a runtime error.

QUESTION 25

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by `//line 2` ?

A. `A 22`
B. `A 11`
C. `A 33`
D. Output cannot be determined until runtime
E. This is no output due to a runtime error.

```
class A{

  protected int y;
  int x;

  public A(int a,int b){
     y=a;x=b;
  }

  public int add(int k){
     return k*(x+y);
  }

  public String toString(){
     return"A "+add(1);
  }

}

class B extends A{

  private static int z;

  public B(int a,int b){
     <1*>
     z++;
  }

  public int add(int k){
     return (z+k)*(x+y);
  }

}
/////////////client code/////////////
A a=new <2*>
out.println(a.add(3));  //line 1
A b=new B(5,6);
out.println(b);  //line 2
```

QUESTION 26

Which of the following could be added to interface `C` so that both classes `A` and `B` could implement it in their current state?

A. `public int add(int k);`
B. `public String toString();`
C. `public int x ;`
D. All of the above.
E. None of the above.

```
interface C{

}
```

QUESTION 27

Which of the following can read a newline characer?

A. `file.nextLine();`

B. `file.nextInt();`

C. `file.next();`

D. A and C only

E. This is no output due to a syntax error.

QUESTION 28

What is output by the line marked `//q28`?

A. `-11`

B. `-10`

C. `-13`

D. `-14`

E. `0`

QUESTION 29

What is output by the line marked `//q29`?

A. `0`

B. `2`

C. `-1`

D. `1`

E. `-2`

QUESTION 30

What is output by the line marked `//q30`?

A. `-32`

B. `-85`

C. `-33`

D. `-84`

E. `0`

```
public int recur(int s){
  if(s<=0)return s;
  return recur(s-2)+recur(s-1)+(s-3);
}

///////client code//////////////////
System.out.println(recur(4));  //q28
System.out.println(recur(-2)); //q29
System.out.println(recur(7));  //q30
```

QUESTION 31

What is the bit storage in the primitive type long?

A. `64` B. `8` C. `4` D. `16` E. `32`

QUESTION 32

What is the output by the line marked `//q32` in the code to the right?

A. `1`

B. `null`

C. `2`

D. There is no output due to a runtime error.

E. There is no output due to a compile error.

QUESTION 33

What is the output by the code to the right ?

A. `[1, 3, 4]` B. `[4, 3, 1]` C. `[4, 3, 2]` D. `[2, 3, 4]`

E. There is no output due to a compile error.

```
LinkedList<String>a;
a=new LinkedList<String>();
a.add("1");
a.add("2");
out.println(a.remove());//q32
a.add("3");
a.add("4");
out.println(a);  //q33
```

QUESTION 34

What is output by the code to the right?

A. `[-1, 3, 8, 4, 212, 9]`
B. `[-1, 3, 4, 8, 9, 212]`
C. `[4, 3, 8, -1, 212, 9]`
D. There is no output due to a runtime error.
E. There is no output due to a compile error.

```
PriorityQueue<Integer> pq;
pq=new PriorityQueue<Integer>();
pq.add(4);
pq.add(3);
pq.add(8);
pq.add(-1);
pq.add(212);
pq.add(9);
System.out.println(pq);
```

QUESTION 35

What data structure is implemented in the code to the right ?

A. max-heap
B. binary search tree
C. queue
D. min-heap
E. stack

QUESTION 36

What is output by the code to the right ?

A. `255`
B. `254`
C. `127`
D. `62`
E. There is no output due to a compile error.

```
System.out.println(127|45&61^152);
```

QUESTION 37

What is the value of the postfix equation to the right ?

A. `-8`
B. `8`
C. `320`
D. `-320`
E. This is no output due to a syntax error.

```
2 3 ^ 8 * 4 5 7 + - /
```

QUESTION 38

What is the runtime of adding all the elements from array `g` of integers to a `TreeSet`, if N is `g.length`?

A. `O(nlogn)` B. `O(n^2)` C. `O(n)` D. `O(logn)` E. `O(n^2logn)`

QUESTION 39

What regex String would split the following String s into the array `{"1234", "1001"}`?

```
String s="1234ABCD.&%#+abcd1011";
```

QUESTION 40

What is the post-order traversal of the binary tree created when the following values are inserted in order ?

`1, 212, -6, 7, 9, -212, 8, -5, -4, 12`

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge Meet #1
Computer Science - KEY

1) B
2) B
3) A
4) D
5) D
6) E
7) A
8) D
9) A
10) A

11) B
12) C
13) D
14) E
15) C
16) D
17) D
18) B
19) A
20) E

21) C
22) A
23) B
24) B
25) A
26) A
27) A
28) B
29) E
30) C

31) A
32) A
33) D
34) A
35) D
36) A
37) A
38) A
39) \\D+,[^0-9]+,[A-Za-z\\.&%#]+
40) -4 -5 -212 -6 8 12 9 7 212 1

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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